

An Introduction to



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Resources

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Free Online Book: Hands-On Programming with R

<https://rstudio-education.github.io/hopr/index.html>

Rstudio IDE Cheatsheet

[RStudio IDE :: Cheatsheet](#)

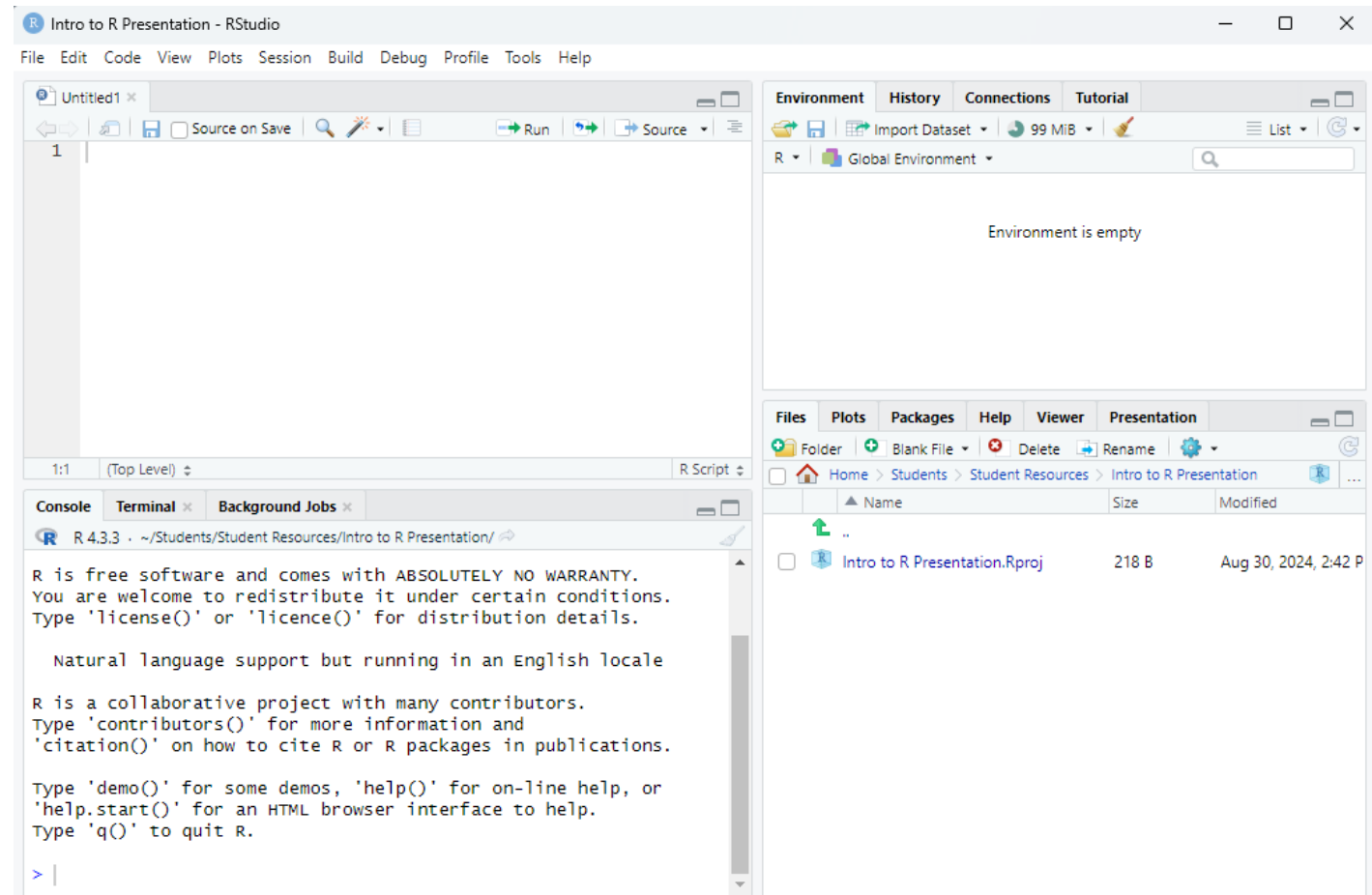
Code and Data used in today's examples and other helpful resources

What is R?

- **R is an Open-Source Software for Statistical Analysis +**
 - An important part of **Open Science** and **Reproducible Research!**
 - Freely available
- **R Foundation** is a non-profit that maintains R and CRAN
 - Large community of users and developers
 - Allows for early adoption of cutting-edge statistics by loading add-on packages (more than 10,000 available!!!)
- **R is also a programming language**
 - Object based

What is RStudio?

- **RStudio is an Integrated Development Environment (IDE)**
 - **RStudio** is also a **GUI** that makes R look the same on all operating systems
 - **Allows you to:**
 - Browse objects
 - View file folders
 - View installed packages
 - Write and edit code
 - View output, charts, and graphs



Let's Dive In!

Some tips:

- Don't be nervous. You're not going to break anything (permanently). **Do not fear the red text!**
 - What to do if you get red text** (in this order):
 1. Read it! Sometimes the solution is right there in the error message.
 2. Google it! **The best coders are great Googlers.**
 3. Ask your peers for help!
 4. E-mail me!
- Make sure all your data, code files, and project file are in the same folder on your computer
- Always save your code!
 - To make sure your results are reproducible
 - To make sure you can go back to code that runs if you break things.
 - To save time in future projects – **Reuse old code when you can!!**

Bonus: R Markdown

- **Make reproducible documents** with tables and figures in R!

Quick Example if we have time

Thank you!



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Reach out if you have questions or need help!
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